

Pre-Registered Falsification Targets for Bubble Spacetime Theory

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Purpose

This document pre-registers specific, falsifiable predictions of Bubble Spacetime Theory (BST) before experimental results are available. BST derives physical constants from the geometry $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ using five integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{max}=137$) and zero free parameters.

We commit to the following predictions, ordered by falsification strength. If any is contradicted by measurement at the stated precision, BST is falsified at tree level.

Prediction 1: No Neutrinoless Double Beta Decay at Any Scale

Claim: BST predicts Dirac neutrinos (Z_3 on CP^2 fiber, Hopf structure forces Dirac). Neutrinoless double beta decay (0vbb) will NOT be observed at any half-life, regardless of detector sensitivity.

Falsification condition: Observation of 0vbb at any half-life above 10^{24} years (current sensitivity $\sim 10^{26}$ years, next generation targets 10^{28} years).

Timeline: LEGEND-200 (~ 2027), nEXO (~ 2030), CUPID (~ 2028).

BST basis: T319 (permanent alphabet), Hopf fiber argument — neutrino mass structure is Dirac, Majorana mass term topologically forbidden by D_{IV}^5 fiber structure.

Strength: Strong. This is a binary prediction (yes/no) with no adjustable threshold. Every null result strengthens BST; a single detection kills it.

Prediction 2: Three Generations Exactly

Claim: $N_c = 3$ forces exactly three fermion generations. No fourth-generation fermion exists at any mass.

Falsification condition: Discovery of a fourth-generation charged lepton or quark at any collider energy.

BST basis: Z_3 symmetry of color sector is exhaustive. $N_c = 3$ is a topological invariant, not a parameter.

Strength: Strong. Binary, mass-independent, collider-testable. The Z_3 constraint is topological and cannot be deformed.

Prediction 3: W Boson Mass = 80.361 GeV

Claim: $m_W = n_C * m_p / (8 * \alpha) = 80.361 \text{ GeV}$.

Falsification condition: Combined LHC measurement gives m_W outside [80.355, 80.367] GeV at 3-sigma confidence. (Window = $\pm 6 \text{ MeV}$, matching the anticipated LHC combined precision of $\sim 5 \text{ MeV}$.)

BST basis: Electroweak sector of D_{IV}^5 . See BST_W_Boson_Precommitment.md.

Strength: Medium. The window is tight (75 ppm) and matches experimental capabilities by ~ 2027 .

Prediction 4: CMB Spectral Index

Claim: $n_s = 1 - n_C/N_{\max} = 1 - 5/137 = 0.96350$.

Falsification condition: Planck/LiteBIRD combined measurement gives n_s outside [0.9615, 0.9655] at 3-sigma. (Window = ± 0.002 , matching the anticipated LiteBIRD sensitivity.)

Current status: Planck 2018 gives $n_s = 0.9649 \pm 0.0042$. BST prediction is 0.3-sigma from central value.

BST basis: Cascade fingerprint of D_{IV}^5 on CMB (Toy 1401).

Strength: Medium. The window is narrower than the current experimental uncertainty, so this prediction is genuinely at risk from improved CMB measurements.

Prediction 5: BaTiO₃ 137-Plane Casimir Peak (Exploratory)

Claim: A Casimir force measurement in a BaTiO₃ wedge cavity will show an anomalous peak when the gap equals exactly 137 lattice planes (54.9 nm for the a-axis, $137 \times 4.009 \text{ \AA}$).

Falsification condition: No anomalous feature within ± 5 planes of the 137-plane spacing, at measurement sensitivity sufficient to resolve 5% modulation of the Casimir force.

Estimated cost: \$25,000 (wedge cavity + PFM measurement).

BST basis: $N_{\text{max}} = 137$ is the spectral cap of D_{IV}^5 . Boundary conditions at 137 lattice planes resonate with the complete eigenvalue spectrum.

Strength: Exploratory. The prediction mechanism (spectral resonance at N_{max} lattice planes) is the least rigorous of the five. A null result would indicate that the spectral cap manifests differently in condensed matter than predicted, without necessarily falsifying the core theory. We include it because it is the cheapest direct experiment (\$25K) and success would be dramatic.

Standing Commitment

All predictions above are parameter-free. None can be adjusted after the fact. If any single prediction among P1-P4 is falsified at the stated precision, the corresponding BST derivation chain is wrong and must be publicly retracted. P5 is classified as exploratory: a null result requires reexamination of the condensed-matter application but does not invalidate BST's particle physics predictions.

Casey Koons. *Bubble Spacetime Theory*. [DATE] Repository: [github.com/\[repo\]](https://github.com/[repo]).
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